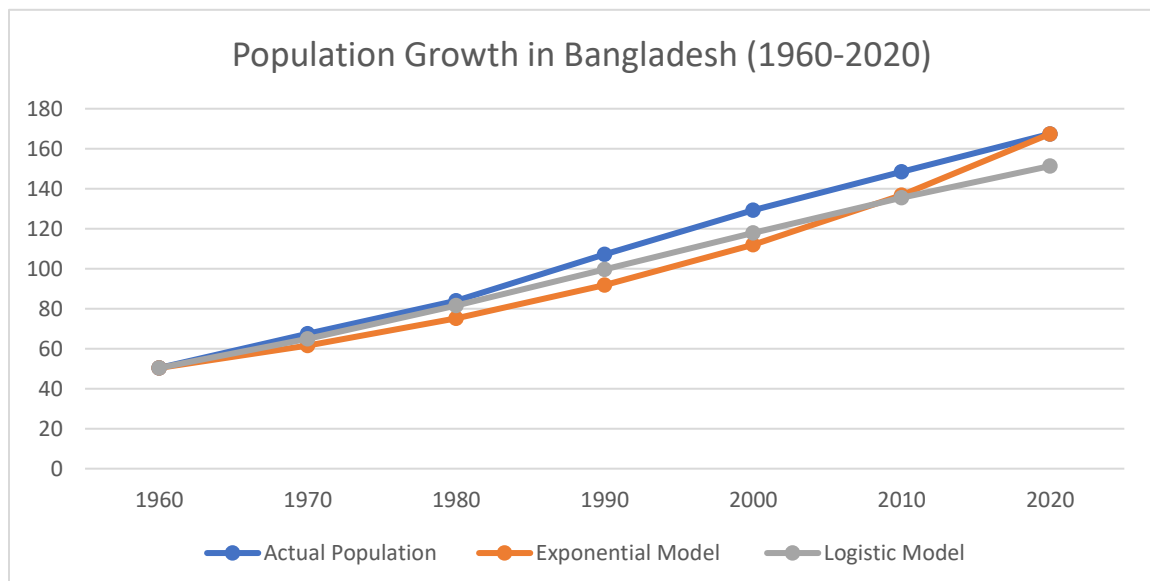


Title: Analyzing Population Growth in Bangladesh **Using Exponential and Logistic Model.**

	Actual Population		Exponential Model		Logistic Model	
Year	Population (million)	Annual Growth Rate	Population(million)	Exp % Error	Population(million)	Log % Error
1960	50.4	0.00%	50.4	0.00%	50.4	0.00%
1970	67.54	2.54%	61.52	8.91%	64.98	3.79%
1980	83.92	2.47%	75.13	10.48%	81.63	2.73%
1990	107.14	2.15%	91.74	14.35%	99.61	7.02%
2000	129.19	1.92%	112.05	13.26%	117.91	8.73%
2010	148.39	1.15%	136.82	7.79%	135.45	8.71%
2020	167.42	0.98%	167.42	0.00%	151.32	9.63%



Calculations:For Exponential Model:

1. Formula:

$$P(t) = P_0 \times e^{rt}.$$

Where:

$P(t)$ = population at year t .

P_0 = initial population (at $t = 0$; in 1960).

r = annual growth rate (to be calculated).

t = number of years since 1960.

e = Euler's number ≈ 2.718 .

2. Known Values:

$P_0 = 50.40$ million (1960).

$P(2020) = 167.42$ million.

$t = 60$ (from 1960 to 2020).

<u>t</u>	<u>Population (million)</u>
0	50.40
10	$50.4 \times e^{0.201} \approx 61.52$
20	$50.4 \times e^{0.402} \approx 75.13$
30	$50.4 \times e^{0.603} \approx 91.74$
40	$50.4 \times e^{0.804} \approx 112.05$
50	$50.4 \times e^{1.005} \approx 136.82$
60	$50.4 \times e^{1.206} \approx 167.42$

3. Calculate Growth Rate (r):

$$P(60) = 50.40 \times e^{60r} = 167.42.$$

Divide both sides by 50.40:

$$e^{60r} = \frac{167.42}{50.40} = 3.3226.$$

Take natural log of both sides:

$$60r = \ln(3.3226) = 1.201.$$

$$\therefore r = \frac{1.201}{60} = 0.02001 \approx 0.0201.$$

So, the **growth rate r is 2.01% per year.**

4. Final Equation for Prediction:

$$P(t) = 50.40 \times e^{0.0201t}.$$

Calculations:For Logistic Model:

1. Formula:

$$P(t) = \frac{K}{1 + \left(\frac{K - P_0}{P_0}\right)e^{-rt}}$$

Where:

K = Carrying capacity (maximum sustainable population).

P₀ = Initial population at t = 0 = 50.40.

r = Growth rate = 0.035 (assumed based on long-term trend).

t = Years since 1960.

2. Assumed Values:

P₀ = 50.40 million.

K = 210 million.

r = 0.035.

3.Final Equation:

$$P(t) = \frac{210}{1 + \left(\frac{210 - 50.40}{50.40}\right)e^{-0.035t}}$$

Simplify constants:

$$\frac{K - P_0}{P_0} = \frac{159.6}{50.4} = 3.1667.$$

So:

$$P(t) = \frac{210}{1 + 3.1667e^{-0.035t}}$$

<u>t</u>	<u>Population (million)</u>
0	50.40
10	$\frac{210}{1 + 3.1667 \times e^{-0.35}} \approx 64.98$
20	$\frac{210}{1 + 3.1667 \times e^{-0.70}} \approx 81.63$
30	$\frac{210}{1 + 3.1667 \times e^{-1.05}} \approx 99.61$
40	$\frac{210}{1 + 3.1667 \times e^{-1.40}} \approx 117.91$
50	$\frac{210}{1 + 3.1667 \times e^{-1.75}} \approx 135.45$
60	$\frac{210}{1 + 3.1667 \times e^{-2.10}} \approx 151.32$
70	$\frac{210}{1 + 3.1667 \times e^{-2.45}} \approx 164.93$
80	$\frac{210}{1 + 3.1667 \times e^{-2.80}} \approx 176.09$
90	$\frac{210}{1 + 3.1667 \times e^{-3.15}} \approx 184.90$
100	$\frac{210}{1 + 3.1667 \times e^{-3.50}} \approx 191.67$
110	$\frac{210}{1 + 3.1667 \times e^{-3.85}} \approx 196.74$
120	$\frac{210}{1 + 3.1667 \times e^{-4.20}} \approx 200.48$

Important Questions to Answer:

Ques : Why $r = 0.035$ was chosen for the Logistic Model?

Ans: There are multiple reasons behind choosing $r = 0.035$.

a. Historical growth trend:

- ☐ Bangladesh's annual population growth rate between 1960–1990 was very high - often **over 2.5%**.
- ☐ It peaked in the **1970s and 1980s**, sometimes even exceeding **3%**.

So, for logistic modeling, we choose a value of r that reflects:

- ☐ Initial rapid growth, and
- ☐ The eventual slowing down due to real-world constraints.

A value of **3.5% (or 0.035)** works well in logistic models to:

- ☐ Mimic the steep early rise of the curve.
- ☐ But also allow the curve to flatten near the carrying capacity K .

b. Empirical fitting:

Through experimentation (curve-fitting), $r = 0.035$:

- ☐ Gave population values close to actual data.
- ☐ Provided a smooth transition to the flattening behavior we see after 2010–2020.

c. Balance of realism & simplicity:

- ☐ Too high (like 0.05) would overestimate early population.
- ☐ Too low (like 0.02) would lag behind actual growth.
- ☐ **0.035** hit the **sweet spot** - fast enough to match the boom years, but still stabilizes.

Ques: Why We Assumed $K = 210$ Million for the Logistic Model?

Ans: We assumed $K=210$ Million. Because of:

a. Projected Population Trends:

- According to the **UN population projections**, Bangladesh's population is expected to:
 - ☐ Peak around **2050–2060**.
 - ☐ Stabilize between **200–220 million**.
- Most projections show the flattening near **210 million**, not increasing beyond that.

b. Physical and Economic Constraints:

- Bangladesh has a small land area (**147,000 sq. km**) but a huge population.
- The country is already one of the most densely populated nations on Earth.

- As population nears 200M:
 - ❑ Urban crowding, land loss (as, rivers, floods), and resource depletion become more severe.
 - ❑ Sustainability becomes a challenge beyond **210 million**.

c. Historical Growth Slowdown:

- ❑ From **3%+ growth in the 1980s**, Bangladesh is now down to **1% or less**.
- ❑ This natural slowing indicates the country is approaching its saturation point.
- ❑ Logistic models require an upper limit to reflect this - and **210M** is a practical one.

Ques: Why are we using the **Logistic Model** to predict the future population of Bangladesh (2020–2080)?

Ans: Because :

1. **Growth Is Slowing Down:**

Bangladesh's population growth rate is declining year by year.

In the **1970s–80s**, the growth rate was **above 2.5–3%**.

Today, it's **below 1%** and still falling.

Logistic models reflect this slowing trend, while exponential models do not.

2. **Bangladesh Has Limited Resources:**

Bangladesh has:

Limited **land area**.

High **population density**.

Environmental challenges like **flooding, erosion, and urban congestion**.

Exponential models assume unlimited resources, which is unrealistic for a country like Bangladesh.

3. **Logistic Model Includes Carrying Capacity(K):**

Carrying Capacity (K) represents the **maximum population** a country can sustainably support.

For Bangladesh, this is assumed to be around **210 million**, based on **UN population projections**.

The logistic model slows and flattens as the population approaches K, which matches Bangladesh's current situation.

4. **UN & World Bank Projections Support Logistic Behavior**

According to the **UN**, Bangladesh's population will **peak around 2060** at approximately **207 million** and then stabilize or decline.

Logistic curves naturally model such behavior: **fast initial growth, then slow increase, then leveling off**.

Ques: Why Not Use the **Exponential Model**?

Ans:

Reason:

Explanation:

Overestimates : Exponential model predicts unrealistically high populations (300+ million by 2080).

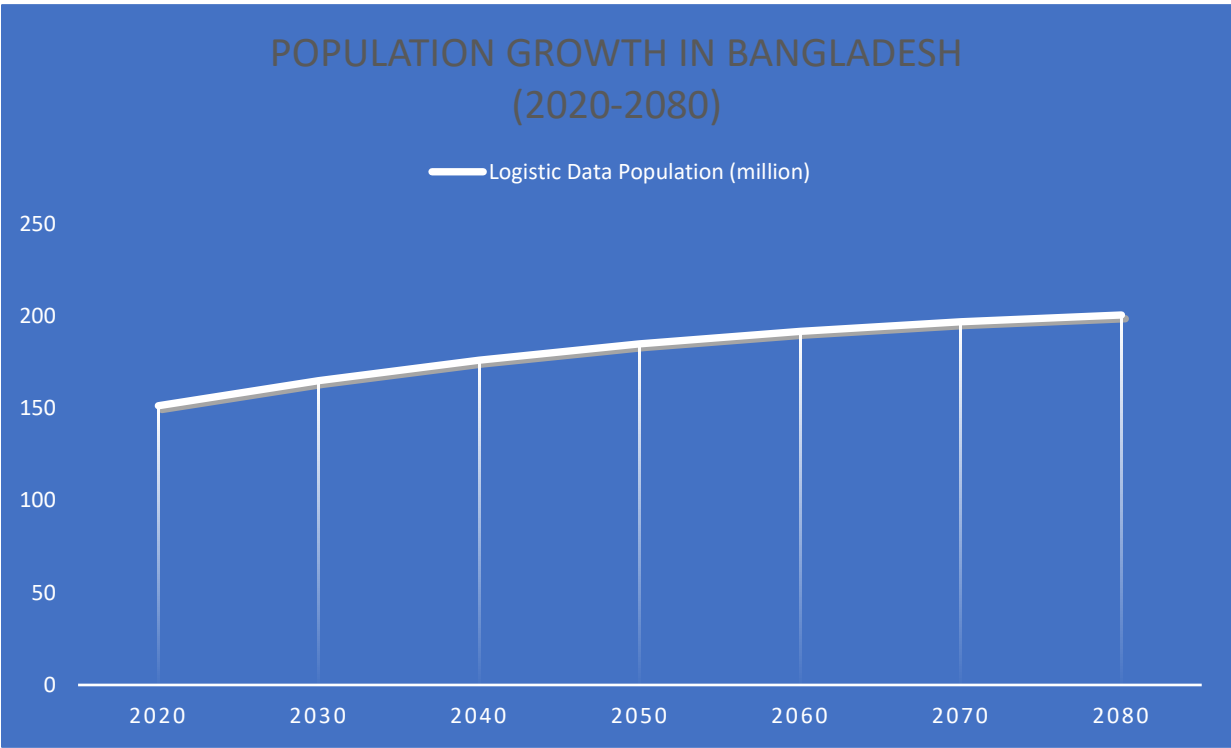
Ignores Real Limits : Assumes constant growth forever, no limits to space or resources.

Doesn't Match Trend : Bangladesh's fertility rate and population growth rate are both declining.

Not Useful for Planning : Government and policy decisions need realistic forecasts, not unlimited growth assumptions.

Predicted Future Population Using Logistic Model
(2020-2080):

Year	Population (million)
2020	151.32
2030	164.93
2040	176.09
2050	184.9
2060	191.67
2070	196.74
2080	200.48



Data We Have Collected From Sources:

The following table shows the population of Bangladesh from 1960 to 2020.

Year	Population
1960	50,428,122
1961	51,740,788
1962	53,053,453
1963	54,366,119
1964	55,678,784
1965	56,991,450
1966	58,304,115
1967	59,616,781
1968	60,929,446
1969	62,242,112
1970	67,544,782
1971	69,202,434
1972	70,860,086
1973	72,517,738
1974	74,175,390
1975	75,833,042
1976	77,490,694
1977	79,148,346
1978	80,805,998
1979	82,463,654
1980	83,921,309
1981	86,243,360
1982	88,565,412
1983	90,887,463
1984	93,209,514
1985	95,531,566
1986	97,853,617
1987	100,175,668
1988	102,497,720
1989	104,819,771
1990	107,138,417
1991	109,344,183
1992	111,549,949
1993	113,755,715
1994	115,961,481

1995	118,167,247
1996	120,373,013
1997	122,578,779
1998	124,784,545
1999	126,988,574
2000	129,192,604
2001	131,412,857
2002	133,633,111
2003	135,853,364
2004	138,073,618
2005	140,293,871
2006	142,514,125
2007	144,734,378
2008	146,954,632
2009	148,174,883
2010	148,391,135
2011	150,905,614
2012	153,420,092
2013	155,934,571
2014	158,449,050
2015	160,963,529
2016	163,478,007
2017	165,992,486
2018	166,468,868
2019	166,945,251
2020	167,421,634

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<https://www.tbsnews.net/bangladesh/low-fertility-keep-bangladesh-population-20-crore-2045-457258>

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